

(Label) Claim	Important Tweets	#Tweets
(UNVERIFIED) Surprising number of vegetarians secretly eat meat	1 @HuffingtonPost then they aren't vegetarians.	33
	2 @HuffingtonPost this article is stupid. If they ever eat meat, they are not vegetarian.	
	3 @HuffingtonPost @laurenisaslayer LOL this could be a The Onion article	
(TRUE) Officials took away this Halloween decoration after reports of it being a real suicide victim. It is still unknown. URL	1 @NotExplained how can it be unknown if the officials took it down.....	46
	2 @NotExplained did anyone try walking up to it to see if it was real or fake? this one seems like an easy case to solve	
	3 @NotExplained thats from neighbours	
(FALSE) CTV News confirms that Canadian authorities have provided US authorities with the name Michael Zehaf-Bibeau in connection to Ottawa shooting	1 @inky_mark @CP24 as part of a co-op criminal investigation one would URL doesn't need facts to write stories it appears.	5
	2 @CP24 I think that soldiers should be armed and wear protective vests when they are on guard any where.	
	3 @CP24 That name should not be mentioned again.	

Table 5: Samples of tweet level explanation for each claim. We sort tweets based on the number of times it was identified as the most relevant tweet to the most important tweet and show the top three tweets. The right most column gives the number of tweets in the thread.

set from another event. To verify our hypothesis, we trained and tested the StA-HiTPLAN + time-delay model by splitting the train and test set randomly. As seen in Table ??, we were able to obtain an F-score of 77.4. (37.5 higher compared to using events split). Because splitting by events a more realistic setting, we would explore methods to make our model event agnostic in future works.

5.5 Explaining the predictions

One key advantage of our model is that we could examine the attention weights in our model to interpret the results of our model. We illustrate how we can generate both token-level and tweet-level explanations for our predictions.

Post-Level Explanations As described in Section 4, we used the self-attention mechanism to aggregate information among tweets. The amount of information a tweet is propagating to another tweet is weighted by the relatedness between the pair, measured by the self-attention weight between them - higher self-attention weight imply higher relatedness. We then use the attention mechanism to interpolate tweets in the final layer before the prediction layer. This generates a representation vector for the claim that would be used for prediction. The attention weight for each tweet indicates the level of importance the model has placed on that tweet for prediction. Higher attention weight implies higher importance. Therefore, to generate the explanations for each claim, we obtain the tweet with the highest attention weight at the final layer - this would give us the most important tweet, $tweet_{impt}$, for prediction. After which, we obtained the most relevant tweet, $tweet_{rel,i}$ to $tweet_{impt}$ at the i^{th} MHA layer. We do so by obtaining tweets with highest self-attention weight with this particular tweet at each MHA layer. The same tweet could be identified as the most relevant tweet multiple times. We ranked each tweet based on the number of times it was identified as a relevant tweet. The top three tweets would be the explanation for this particular claim. Table 5 shows examples of the top three tweets: we see replies that cast doubt on, or confirm, the claim. These replies were accurately identified and used by our model in

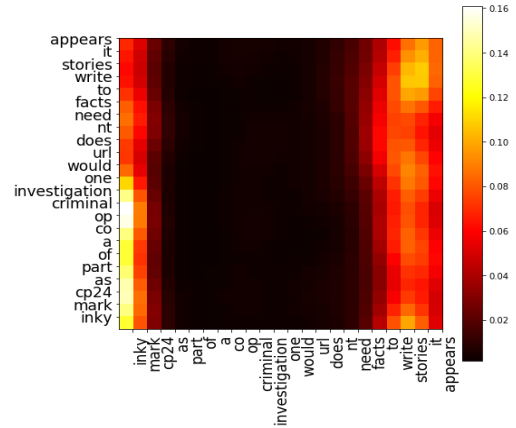


Figure 3: Heatmap showing the important tokens with token-level self-attention for a fake claim in PHEME. A lighter colour means higher importance.

debunking or confirming the factuality of claims despite a large number of other tweets present in the conversation.

Token-Level Explanation As described in Section 5.5, we could extract the important tweets to a prediction - These tweets provide tweet-level explanations for our model. Following the steps described, we obtained "*@inky_mark @CP24 as part of a co-op criminal investigation one would URL doesn't need facts to write stories it appears.*" as the most important tweet to predict the fake claim, "*CTV News confirms that Canadian authorities have provided US authorities with the name Michael Zehaf-Bibeau in connection to Ottawa shooting*" correctly. This claim was obtained from the PHEME dataset. As shown in Figure 3, most tokens in the tweet have equally high attention weights with tokens at the end of the tweet. A further examination of the attention weights show that high weights were placed on the phrase "facts to write stories it appears". As such, "facts to write stories it appears" could be deemed as an important phrase that could explain the prediction of our model for this claim.

6 Conclusion

We have proposed three models that outperformed state-of-the-art models on three data sets. Our model utilizes the self-attention mechanism to model pairwise interactions between posts. We utilized the attention mechanism to provide possible explanations of the prediction by extracting the important posts that resulted in the prediction. We also investigated mechanisms to capture structure and time information.

In this paper, we focused only on data with community response. Recent papers that perform rumor detection with user identity information have shown superior results. Lastly, another direction in fake news detection is fact checking with reliable sources. Fact checking and rumor detection could provide complementary information and could be done in a joint manner. We would consider this in future.

Acknowledgement

We would like to thank all the anonymous reviewers for their help and insightful comments. This piece of work was done when Qian Zhong was at SMU.

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